le.chang [at] umontreal [dot] ca
Permanent Resident of Canada
Google Scholar
0000-0002-0966-7923
ResearchGate
le-chang.github.io

Le **Chang**, PhD

CIHR Postdoctoral Research Fellow in computational genetics and bioinformatics.
Gagliano Taliun Lab, Research Centre, Montreal Heart Institute, 5000 rue Bélanger, Montreal, Quebec H1T 1C8, Canada

Academic Positions

Sept. 2024–*present* **Montreal Heart Institute & Université de Montréal** *Montreal, Canada*
Postdoctoral Research Fellow, Research Centre (MHI) and Faculty of Medicine (UdeM)

Sept. 2023–Aug. 2024 **UTHealth Houston** *Houston, USA*
Postdoctoral Research Fellow, Center for Precision Health, McWilliams School of Biomedical Informatics

Education and Training

May 2023 **McGill University, Montreal, Canada**
PhD, Human Genetics (Bioinformatics), Department of Human Genetics
- Supervisor: Jianguo (Jeff) Xia · NSERC CREATE MATRIX Scholarship
- Dissertation: Network-based approaches towards understanding genetics of metabolism and regulatory interactions

May 2019 **Queen's University, Kingston, Canada**
MBI, Biomedical Informatics, School of Computing and the Department of Biomedical and Molecular Sciences
- Supervisor: Qingling Duan, Computational Genomics Laboratory
- Project: Genome-wide association study of human milk oligosaccharides in the CHILD Cohort Study

June 2015 **University of Alberta, Edmonton, Canada**
BSc, Nutrition and Food Science, Faculty of Agricultural, Life & Environmental Sciences
- Graduated with Distinction; Dean's List and First-Class Standing

June 2015 **Northwest A&F University, Yangling, China**
BEng, Bioengineering
- "2+2" dual-degree program with the University of Alberta

Professional Experience

June 2015–April 2018 **Maple Leaf Foods Inc.** *Canada*
Leadership Trainee
Hired into the company's leadership trainee program; progressed through Production Supervisor (2015–2016) and Business Process Analyst (2016–2017) to Supply Planning Analyst (2017–2018). Demand forecasting and inventory analytics; managed the recipe database for 11 manufacturing facilities; supervised a 20-person production team.

Funding

2025–2027 **CIHR Fellowship**, Canadian Institutes of Health Research: "Unraveling the genetic architecture and causal pathways of hematuria: a comprehensive multi-ancestry meta-analysis" (my role: **principal applicant**; $140,000)

2025–2026 **Postdoctoral Fellowship Award**, Heart and Stroke Foundation of Canada (my role: **principal applicant**; $30,000; awarded and declined in favour of the CIHR Fellowship)

2024 **CPRIT BIG-TCR Postdoctoral Fellowship**, Cancer Prevention and Research Institute of Texas, Biomedical Informatics, Genomics and Translational Cancer Research Training Program: "Decoding the role of viral transcriptional regulators in cancer" (my role: **principal applicant**; USD $52,704)

2025–*present* **Approved data access**, Canadian Longitudinal Study on Aging, application 2501015: "Multi-omics Mendelian randomization approaches to identify causal modifiable risk factors for dementia" (three-year access; principal applicant: Sarah A. Gagliano Taliun; my role: **lead analyst**)

Scholarships and Awards

2025–2027 **CIHR Fellowship**, Canadian Institutes of Health Research

2025–2026 **Postdoctoral Fellowship Award**, Heart and Stroke Foundation of Canada (declined)

2024 **CPRIT BIG-TCR Postdoctoral Fellowship**

Sept. 2022 **MANA Early-Career Travel Award**, Metabolomics Association of North America

June 2020 **ISMB 2020 Fellowship Award**, International Society for Computational Biology

2019–2023 **NSERC CREATE MATRIX Scholarship**, NSERC CREATE MATRIX Training Program

May 2019 **Graduate Excellence Award**, Department of Human Genetics, McGill University

Jan. 2019 **AllerGen Conference Travel Award**, AllerGen NCE Inc.

June 2015 **Graduation with Distinction**, University of Alberta

2014, 2015 **Dean's List and First-Class Standing**, University of Alberta

May 2015 **Ronald A. Chisholm Undergraduate Scholarship**, Canadian Meat Science Association

Jan. 2015 **Ian Morrison Undergraduate Leadership Award**, University of Alberta

Jan. 2015 **Macgregor Smith Public Speaking Competition**, Third Place, University of Alberta

Oct. 2014 **Mary (Davidson) Howard Bar None Award**, University of Alberta

Aug. 2014 **Student Challenge Competition**, First Place Team, Canadian Institute of Food Science and Technology

Feb. 2014 **Hilda Wilson Memorial Volunteer Recognition Award**, University of Alberta

May 2013 **NWAFU–ALES Undergraduate Transfer Program Award**, University of Alberta

Publications (n = 14; * denotes equal contribution)

Summary Since 2019: 11 peer-reviewed journal articles (6 as (co-)first author, 5 as co-author), 2 book chapters (first author) and 1 published conference abstract (first author). Three ESI Highly Cited Papers (Web of Science); miRNet 2.0 has been cited more than 900 times.

Peer-reviewed journal articles

[J1] **Genome-wide cross-trait analysis to compare genetic architecture between Parkinson's disease and kidney-related traits**
L. Chang*, S. Gawhary*, L. Maameri, W. Belbellaj, F. Lona-Durazo, S.A. Gagliano Taliun. *Nature Communications*, 2026. Published 18 September 2026. DOI: 10.1038/s41467-026-77920-3

[J2] **Exploring and visualizing stratified genome-wide association study results with PheWeb 2**
J. Bellavance*, H. Xiao*, **L. Chang**, M. Kazemi, S. Wickramasinghe, A.J. Mayhew, P. Raina, P. VandeHaar, D. Taliun, S.A. Gagliano Taliun. *Nature Genetics*, 2026; 58(2):233–235. DOI: 10.1038/s41588-025-02469-8. PMID: 41555088

[J3] **The relationship between kidney health and neurodegenerative diseases**
M. Zuo*, **L. Chang***, N. Neale, L. Maameri, S. Gawhary, F. Lona-Durazo, S.A. Gagliano Taliun. *Brain*, 2025; 148(8):2616–2630. DOI: 10.1093/brain/awaf113. PMID: 40120077. Podcast: open.spotify.com/episode/0OdP5vSjzd7nMwLEeetqsR

[J4] **Identification and catalog of viral transcriptional regulators in human diseases**
C. Citu, **L. Chang**, A.M. Manuel, N. Enduru, Z. Zhao. *iScience*, 2025; 28(3):112081. DOI: 10.1016/j.isci.2025.112081

[J5] **Human milk oligosaccharides are associated with maternal genetics and respiratory health of human milk-fed children**
A. Ambalavanan, **L. Chang**, J. Choi, Y. Zhang, S.A. Stickley, Z.Y. Fang, K. Miliku, B. Robertson, C. Yonemitsu, S.E. Turvey, P.J. Mandhane, E. Simons, T.J. Moraes, S.S. Anand, G. Paré, J.E. Williams, B.M. Murdoch, G.E. Otoo, S. Mbugua, E.W. Kamau-Mbuthia, E.W. Kamundia, D.K. Gindola, J.M. Rodriguez, R.G. Pareja, D.W. Sellen, S.E. Moore, A.M. Prentice, J.A. Foster, L.J. Kvist, H.L. Neibergs, M.A. McGuire, M.K. McGuire, C.L. Meehan, M.R. Sears, P. Subbarao, M.B. Azad, L. Bode, Q. Duan. *Nature Communications*, 2024; 15:7735. DOI: 10.1038/s41467-024-51743-6

[J6] **A data-centric perspective on exposomics data analysis**
L. Chang*, J. Ewald*, F. Hui, S. Bayen, J. Xia. *Exposome*, 2024; 4(1):osae005. DOI: 10.1093/exposome/osae005

[J7] **mGWAS-Explorer 2.0: causal analysis and interpretation of metabolite–phenotype associations**
L. Chang, G. Zhou, J. Xia. *Metabolites*, 2023; 13(7):826. DOI: 10.3390/metabo13070826

[J8] **mGWAS-Explorer: linking SNPs, genes, metabolites, and diseases for functional insights**
L. Chang, G. Zhou, H. Ou, J. Xia. *Metabolites*, 2022; 12(6):526. DOI: 10.3390/metabo12060526

[J9] **Using MetaboAnalyst 5.0 for LC–HRMS spectra processing, multi-omics integration and covariate adjustment of global metabolomics data**
Z. Pang, G. Zhou, J. Ewald, **L. Chang**, O. Hacariz, N. Basu, J. Xia. *Nature Protocols*, 2022; 17(8):1735–1761. DOI: 10.1038/s41596-022-00710-w. ***ESI Highly Cited Paper***

[J10] **MetaboAnalyst 5.0: narrowing the gap between raw spectra and functional insights**
Z. Pang, J. Chong, G. Zhou, D.A. de Lima Morais, **L. Chang**, M. Barrette, C. Gauthier, P.-É. Jacques, S. Li, J. Xia. *Nucleic Acids Research*, 2021; 49(W1):W388–W396. DOI: 10.1093/nar/gkab382. ***ESI Highly Cited Paper***

[J11] **miRNet 2.0: network-based visual analytics for miRNA functional analysis and systems biology**
L. Chang, G. Zhou, O. Soufan, J. Xia. *Nucleic Acids Research*, 2020; 48(W1):W244–W251. DOI: 10.1093/nar/gkaa467. ***ESI Highly Cited Paper; more than 900 citations***

Book chapters

[B1] **Computational approaches for studying signal transduction networks at cellular context**
L. Chang, J. Xia, Z. Zhao. In: S. Ranganathan, M. Cannataro, A.M. Khan (eds) *Encyclopedia of Bioinformatics and Computational Biology*, 2nd edition. Elsevier, 2025; pp. 436–444.

[B2] **MicroRNA regulatory network analysis using miRNet 2.0**
L. Chang, J. Xia. In: Q. Song, Z. Tao (eds) *Transcription Factor Regulatory Networks. Methods in Molecular Biology.* Springer, 2022; pp. 185–204.

Published conference abstracts

[A1] **Genome-wide association study of human milk oligosaccharides in the CHILD Study**
L. Chang, A. Ambalavanan, A. Lamri, B. Robertson, C. Yonemitsu, S. Turvey, P. Mandhane, A. Becker, T. Moraes, S. Anand, G. Paré, D. Lefebvre, M. Sears, P. Subbarao, L. Bode, M. Azad, Q. Duan. *Allergy, Asthma & Clinical Immunology* (AllerGen 2019 Research Conference abstracts), 2019.

Thesis

Network-based approaches towards understanding genetics of metabolism and regulatory interactions
L. Chang. PhD dissertation, *Department of Human Genetics, McGill University*, 2023.

Manuscripts Under Review

[M1] **Ten simple rules for fostering the next generation of computational biologists**
L. Chang, S.A. Gagliano Taliun. *PLOS Computational Biology*, under review.

Software and Research Resources

miRNet 2.0 (mirnet.ca) *lead developer and first author*
Network-based visual analytics platform for microRNA functional analysis and systems biology, integrating miRNAs, genes, transcription factors, SNPs, lncRNAs and circRNAs across multiple species. Responsible for system architecture, database design, Java/R back end, interactive network visualization, deployment, tutorials and multi-year user support. More than 70,000 users; ESI Highly Cited Paper.

mGWAS-Explorer 1.0 and 2.0 (mgwas.ca) *lead developer and first author*
Platforms linking SNPs, genes, metabolites, diseases and molecular QTL evidence for the interpretation of metabolite–phenotype associations; version 2.0 adds two-sample Mendelian randomization across more than 4,000 metabolites and literature-derived semantic triples, with an accompanying R package.

MetaboAnalyst 5.0 (metaboanalyst.ca) *core contributor*
Designed and implemented the data-driven network analysis module and rebuilt the underlying KEGG interaction database for a platform serving more than 500,000 users.

PheWeb 2 (github.com/GaglianoTaliun-Lab/PheWeb2) *core contributor*
Open-source platform for exploring and visualizing stratified genome-wide association study results.

Analysis pipelines
Reproducible, version-controlled workflows (Nextflow, SLURM) for multi-ancestry GWAS meta-analysis (REGENIE), LD score regression, local genetic correlation (LAVA), conjunctional FDR, colocalization and multivariable Mendelian randomization on Digital Research Alliance of Canada clusters.

Oral Presentations

Oral Presentations as Invited Speaker

26 Dec. 2024 *Deciphering genetic architecture of human metabolism and gene regulatory networks: implications for public health*
International Young Scholars Forum, School of Public Health, Southeast University *Nanjing, China*

21 Dec. 2024 *Deciphering genetic architecture of human metabolism and gene regulatory networks: implications for public health*
International Young Scholars Forum, School of Public Health, Nanjing Medical University *Nanjing, China*

28 Nov. 2024 *Multi-omics data integration in precision medicine: applications and prospects*
International Young Scholars Forum, Institute of Future Agriculture, Northwest A&F University *Virtual meeting*

2023 *Deciphering genetic architecture of human metabolism and regulatory interactions*
Center for Precision Health Seminar in Precision Medicine, University of Texas Health Science Center at Houston *Houston, Texas*

7 Sept. 2022 *Decoding genetic variants of molecular phenotypes and regulatory networks*
Institute of Intelligent Healthcare, School of Artificial Intelligence, Nanjing University of Information Science and Technology *Virtual meeting*

2022 *MicroRNA regulatory network analysis using miRNet 2.0*
McGill University – Jilin University Bioinformatics Workshop *Virtual meeting*

2021 *Human milk oligosaccharides*
Nutraceuticals and Functional Foods Global Classroom Session, Universiti Malaysia Pahang *Virtual meeting*

Oral Presentations

2019 *Investigation of genetic modulators of human milk oligosaccharide concentrations in nursing mothers*
Three Minute Thesis Competition, Queen's University *Kingston, Ontario*

Jan. 2019 *Genome-wide association study of human milk oligosaccharides in the CHILD Study*
AllerGen 2019 Research Conference, trainee oral presentation *Toronto, Ontario*

Poster Presentations (presenting author marked "presenter"; * denotes equal contribution)

Oct. 2026 *Genome-wide cross-trait analysis to compare genetic architecture between Parkinson's disease and kidney-related traits*
L. Chang* (presenter), S. Gawhary*, L. Maameri, W. Belbellaj, F. Lona-Durazo, S.A. Gagliano Taliun
American Society of Human Genetics (ASHG) Annual Meeting *Montreal, Quebec*

Oct. 2026 *PheWeb2.1: Exploring stratified GWAS and interaction results with the ability to easily append new summary statistics*
J. Boulais-Richard, H. Xiao, **L. Chang**, P. Raina, D. Taliun, S.A. Gagliano Taliun
American Society of Human Genetics (ASHG) Annual Meeting *Montreal, Quebec*

Oct. 2026 *Baseline glycemia drives genetic associations with incident type 2 diabetes in the Canadian Longitudinal Study on Aging*
S. Wickramasinghe, J. Boulais-Richard, H. Xiao, **L. Chang**, S.A. Gagliano Taliun, D. Taliun
American Society of Human Genetics (ASHG) Annual Meeting *Montreal, Quebec*

Oct. 2025 *Assessment of global and local genetic correlations provides insights on the kidney–brain axis*
L. Chang* (presenter), S. Gawhary*, L. Maameri, W. Belbellaj, F. Lona-Durazo, S.A. Gagliano Taliun
American Society of Human Genetics (ASHG) Annual Meeting *Boston, Massachusetts*

5 June 2025 *Assessment of global and local genetic correlations between kidney-related traits and late-onset neurodegenerative diseases*
S. Gawhary, **L. Chang**, L. Maameri, W. Belbellaj, F. Lona-Durazo, S.A. Gagliano Taliun
Journée de la Recherche de l'Institut de Cardiologie de Montréal *Montréal, Québec*

5 June 2025 *A review of the relationship between kidney health and neurodegenerative diseases*
M. Zuo, **L. Chang**, N. Neale, L. Maameri, S. Gawhary, F. Lona-Durazo, S.A. Gagliano Taliun
Journée de la Recherche de l'Institut de Cardiologie de Montréal *Montréal, Québec*

27 May 2025 *Does kidney health influence dementia and neurodegeneration*
M. Zuo, **L. Chang**, N. Neale, L. Maameri, S. Gawhary, F. Lona-Durazo, S.A. Gagliano Taliun
VAST Training Program Conference *Toronto, Ontario*

2025 *Principal component analysis on the European subset from the Canadian Longitudinal Study of Aging*
S. Wickramasinghe, H. Xiao, J. Bellavance, **L. Chang**, S.A. Gagliano Taliun, D. Taliun
McGill Undergraduate Science Showcase *Montreal, Quebec*

June 2023 *mGWAS-Explorer 2.0: causal analysis and interpretation of metabolite–phenotype associations*
L. Chang (presenter), G. Zhou, J. Xia
4th Annual Canadian Metabolomics Conference *Niagara-on-the-Lake, Ontario*

Sept. 2022 *mGWAS-Explorer: linking SNPs, genes, metabolites and diseases for functional insights*
L. Chang (presenter), G. Zhou, H. Ou, J. Xia
4th Annual Conference of the Metabolomics Association of North America (MANA) *Edmonton, Alberta*

May 2022 *mGWAS-Explorer: linking SNPs, genes, metabolites and diseases for functional insights*
L. Chang (presenter), G. Zhou, H. Ou, J. Xia
2022 Ottawa Institute of Systems Biology Symposium *Cornwall, Ontario*

June 2021 *mGWAS-Explorer: a visual analytics platform connecting metabolomics with genomics*
L. Chang (presenter), G. Zhou, H. Ou, J. Xia
17th Annual Conference of the Metabolomics Society *Virtual meeting*

June 2021 *MetaboAnalyst 5.0: narrowing the gap between raw spectra and functional insights*
Z. Pang, J. Chong, G. Zhou, D.A. de Lima Morais, **L. Chang**, M. Barrette, C. Gauthier, P.-É. Jacques, S. Li, J. Xia
17th Annual Conference of the Metabolomics Society *Virtual meeting*

July 2020 *miRNet 2.0: network-based visual analytics for miRNA functional analysis and systems biology*
L. Chang (presenter), G. Zhou, O. Soufan, J. Xia
28th Conference on Intelligent Systems for Molecular Biology (ISMB 2020), Network Biology (NetBio) COSI poster session. Video: youtube.com/watch?v=xyIDNN9tuiU *Virtual meeting*

Oct. 2019 *Genome-wide association study of human milk oligosaccharides in lactating mothers and interaction effects on lung health among infants in the CHILD Study*
A. Ambalavanan, **L. Chang**, J. Choi, A. Lamri, B. Robertson, C. Yonemitsu, S.E. Turvey, P.J. Mandhane, A.B. Becker, T.J. Moraes, S.S. Anand, G. Paré, D.L. Lefebvre, M.R. Sears, P. Subbarao, L. Bode, M.B. Azad, Q. Duan
69th Annual Meeting of the American Society of Human Genetics (ASHG), poster 2948 *Houston, Texas*

June 2019 *GWAS of maternal human milk oligosaccharides demonstrates protective effects on preschool wheeze in the CHILD Study*
A. Ambalavanan*, **L. Chang***, J. Choi, A. Lamri, B. Robertson, C. Yonemitsu, S. Turvey, P. Mandhane, A. Becker, T. Moraes, S. Anand, G. Paré, D. Lefebvre, M. Sears, P. Subbarao, L. Bode, M. Azad, Q. Duan
8th Annual Canadian Human and Statistical Genetics Meeting *Montebello, Quebec*

June 2019 *mGWAS-DB: elucidating genetically influenced metabotypes for mGWAS*
L. Chang (presenter), H. Ou, O. Soufan, G. Zhou, J. Xia
8th Annual Canadian Human and Statistical Genetics Meeting *Montebello, Quebec*

Jan. 2019 *Genome-wide association study of human milk oligosaccharides in the CHILD Study*
L. Chang (presenter), A. Ambalavanan, A. Lamri, B. Robertson, C. Yonemitsu, S. Turvey, P. Mandhane, A. Becker, T. Moraes, S. Anand, G. Paré, D. Lefebvre, M. Sears, P. Subbarao, L. Bode, M. Azad, Q. Duan
AllerGen 2019 Research Conference *Toronto, Ontario*

May 2014 *Poultry glycosaminoglycans: a new source of dietary 'animal fibers' and candidate prebiotics?*
L. Chang (presenter), M. Ndagijimana, Z. Khiari, M. Betti
Alberta Poultry Research Centre Annual General Meeting *Edmonton, Alberta*

Teaching

At McGill University — Graduate level

Winter 2023 **BINF 531: Statistical Bioinformatics**
Teaching Assistant
Ran the hands-on tutorials and marked the analyses for this graduate course.

Fall 2021 **BTEC 501: Bioinformatics**
Teaching Assistant
Ran the hands-on tutorials and marked the analyses for this graduate course.

Workshops, tutorials and user training

2022 **McGill University – Jilin University Bioinformatics Workshop** *Montreal*
Instructor
Taught a hands-on miRNet 2.0 session on microRNA regulatory network analysis to trainees from China.

2020–2023 **miRNet and mGWAS-Explorer**
Author of training materials
Online tutorials, documentation, slide decks and video tutorials for the two platforms (audience: more than 70,000 miRNet users), and a step-by-step protocol chapter in *Methods in Molecular Biology* (2022).

2020–2023 **User support**
Answered user questions from students, clinicians and researchers by email and on the laboratory's online user forum; user questions fed back into the tutorials and into new platform features.

Supervision/Mentorship

2025–*present* **Trainee co-supervision** *Montreal Heart Institute / Université de Montréal*
Master's-level researchers — cross-trait genetics of liver markers and neurodegenerative disease (with S.A. Gagliano Taliun)

June–Aug. 2024 **Trainee co-supervision** *University of Texas Health Science Center at Houston*
Undergraduate summer intern, CPRIT BIG-TCR Summer Intern Program — deep-learning model for prediction of HTLV-1 viral integration sites

June 2021–June 2022 **Co-mentoring of undergraduate students** *McGill University*
Two undergraduate research students — community forum and RASA conversational agent for web-based bioinformatics platforms

May 2013–Sept. 2015 **Peer mentoring** *University of Alberta*
Two international undergraduate students — academic guidance and referral to campus support services

Certifications and Other Training

Sept. 2025 Potential and Strengths Coach Course, Freedom Life Coach Limited, ICF CCE

Sept. 2024 CIHR Sex- and Gender-Based Analysis Plus (SGBA+) online training modules, Canadian Institutes of Health Research

June 2020 Informatics and Statistics for Metabolomics, Canadian Bioinformatics Workshops

March 2019 Expanding Horizons Professional Development Certificate, Queen's University

Aug. 2018 Ontario Summer School on High Performance Computing (Compute Ontario)

Oct. 2015 Six Sigma White Belt, Maple Leaf Foods Inc.

March 2015 Competent Communicator, Toastmasters International

Dec. 2014 ULead Leadership Program, University of Alberta

Professional Service

Feb. 2025–Feb. 2028 **Youth Editor**, *iMeta*

2024–*present* **Reviewer** for *Brain Communications*, *Journal of Alzheimer's Disease*, *Scientific Reports* and other journals

Sept. 2024–*present* **Member**, ASHG Volunteer Pool

Nov. 2018 **Adjudicator**, Canadian Undergraduate Conference on Healthcare, Queen's University

Feb. 2014 **Organizer and host**, ALES Networking Career Workshop, Faculty of Agricultural, Life & Environmental Sciences, University of Alberta

2013–2015 **VP Communications**, Nutrition and Food Science Students' Association, University of Alberta

Professional memberships American Society of Human Genetics (2018–present); Metabolomics Association of North America (2022–present); Global Organisation for Bioinformatics Learning, Education & Training, GOBLET (2018–present); European Society of Human Genetics (2018); International Society for Computational Biology (2020–present); International Common Disease Alliance (2019–present); Golden Key International Honour Society (2014–present); AllerGen NCE, student member (2018–2019)

Knowledge Translation and Outreach

June 2025 **Invited guest**, The Brain Podcast (official podcast of the journal *Brain*) — "Nephrology Meets Neurology: How Kidney Health Links to Neurodegeneration"; discussed the kidney–brain findings of the *Brain* 2025 review for clinicians, researchers and the public

Oct. 2014–May 2015 **Presenter**, Fun Speakers Toastmasters Club — public talks on food safety, genetically modified foods and alternative protein sources

Technical Skills

Statistical genetics GWAS (REGENIE, PLINK), multi-ancestry meta-analysis, global genetic correlation (LDSC), local genetic correlation (LAVA), conjunctional FDR, fine-mapping (SuSiE), colocalization, univariable and multivariable Mendelian randomization (TwoSampleMR, MVMR-cML-SuSiE), polygenic scores, sex-stratified and chromosome X analysis, molecular QTL (eQTL, mQTL, pQTL) integration

Multi-omics	Genomics, metabolomics, transcriptomics, proteomics, microRNA regulation, exposomics; network analysis
Programming and infrastructure	R, Python (pandas, scikit-learn, PyTorch), Java, JavaScript, SQL, Bash; full-stack web application development; Nextflow and SLURM pipelines on Digital Research Alliance of Canada clusters; Git; containerized, reproducible workflows
Data resources	UK Biobank, Million Veteran Program, Michigan Genomics Initiative, FinnGen, Canadian Longitudinal Study on Aging, CHILD Cohort Study, GTEx, UKB-PPP

Languages

Chinese (Mandarin; native) · English (full professional proficiency)

Last updated: October 31, 2026